

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years shipping Node.js and TypeScript services backed by MongoDB. Led a zero-downtime migration of 20+ production apps on Parse/MongoDB to Firebase at MTailor, reporting directly to the CTO. Architected a Kafka + Docker + Kubernetes microservices backend for FIDA Uganda. Event-driven systems and high-concurrency transactional APIs are my lane — now applying that to ticketing infrastructure at vivenu.

CORE COMPETENCIES

Node.js / TypeScript

MongoDB

REST API Design

Microservices Architecture

Event-Driven Systems (Kafka, Pub/Sub)

Docker + Kubernetes

Modern JavaScript (ES2015+)

GCP + AWS

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote) — Node.js / TypeScript / MongoDB

- Led end-to-end migration of 20+ production applications from **Parse/MongoDB** to Firebase, zero downtime, reporting directly to the CTO.
- Built real-time two-way sync between **MongoDB and Firestore** with Node.js + Google Pub/Sub; idempotent consumers, back-pressure, schema translation.
- Shipped an Express Shipping feature generating an additional \$40 revenue per order through API design + integration.
- Built a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.
- Saved the company \$5,000/month by migrating all services off AWS to GCP.

MTailor (Contractor)

Mar 2022 – Jul 2022

Contractor — MongoDB → Firestore Migration

- Built a migration script reading MongoDB and writing to Firestore with complex processing logic across millions of records.
- Authored SDK documentation (Node.js + Python) for the engineering team.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built ad-hoc jobs querying over 2 million Firestore records from Morningstar APIs.
- Extended backend REST APIs via Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Legal Tech for NGOs

- Led a team of 4 developers; set architecture and code review culture.
- Architected a microservices backend using **Apache Kafka, Docker, Kubernetes** for FIDA Uganda case management.
- Built a cross-platform React Native app with Firebase push notifications and a USSD service talking to Node.js via gRPC.

PROJECTS

FIDA Uganda Microservices Backend

Multi-actor case management platform decomposed into Kafka-backed services, containerised with Docker, orchestrated on Kubernetes. REST + gRPC interfaces.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC

Real-time MongoDB ↔ Firestore Sync (MTailor)

Live dual-write pipeline supporting 20+ apps during a zero-downtime migration. Idempotent consumers, exactly-once semantics on critical writes.

Node.js · TypeScript · MongoDB · GCP Pub/Sub · Firestore

Express Shipping API (MTailor)

Designed and shipped an additional shipping tier behind a new API endpoint integrated with carrier systems. Generated \$40 additional revenue per order.

Node.js · TypeScript · REST APIs · Firebase

EDUCATION

B.Sc. Computer Engineering — Makerere University

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, TypeScript, MongoDB, SQL, REST APIs, GCP, Firebase, Python, React, Git, Linux

Intermediate: Go, AWS, Rust, React Native, PostgreSQL

Patterns: Microservices, event-driven systems (Kafka, NATS, Pub/Sub), Docker/K8s, zero-downtime migrations

Languages: English · Kiswahili